



File Transfer on OneEdge

Application Note

80654NT11973A Rev. 1 – 2022-06-14

APPLICABILITY TABLE

PRODUCTS	Platform Version ID
ME910C1-AU ME910C1-E1 ME910C1-E2 ME910C1-NA ME910C1-NV ME910C1-WW ML865C1-EA ML865C1-NA	30
ME910G1-W1 ME910G1-WW ME310G1-W1 ME310G1-WW ME310G1-W2 ML865G1-WW	37
LE910C1-EUX LE910C1-SAX LE910C1-SVX	25

Table 1: applicability table

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1. INTRODUCTION

1.1. Scope

This document describes the file transfer through OneEdge environment. All the features and solutions described in this document are applicable to all the variants listed in the applicability table.

1.2. Audience

This document is intended for Telit customers, especially system integrators, about to implement their applications using the Telit modules.

1.3. Contact Information, Support

For general contact, technical support services, technical questions and report of documentation errors contact Telit Technical Support at:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use:

<https://www.telit.com/contact-us/>

For detailed information about where you can buy the Telit modules or for recommendations on accessories and components visit:

<https://www.telit.com>

Our aim is to make this guide as helpful as possible. Keep us informed of your comments and suggestions for improvements.

Telit appreciates the user feedback on our information.

1.4. Symbol Conventions



Danger: This information MUST be followed or catastrophic equipment failure or personal injury may occur.



Warning: Alerts the user on important steps about the module integration.



Note/Tip: Provides advice and suggestions that may be useful when integrating the module.



Electro-static Discharge: Notifies the user to take proper grounding precautions before handling the product.

Table 2: Symbol Conventions

All dates are in ISO 8601 format, that is YYYY-MM-DD.

1.5. Related Documents

- OneEdge Getting Started Guide: 1w0301585 Rev. 8
- AT Command Reference Guide: 80529ST10815A, 80617ST10991A
- LwM2M AT commands User Reference Guide: 80529ST10974A, 80617ST11022A
- OMA Lightweight M2M Specification: OMA-TS-LightweightM2M-V1_0_2-20180209-A
- OneEdge Firmware Management App: 80654NT11889A
- Telit IP Easy User Guide: 80000ST10028A

2. OVERVIEW

The possibility to update remotely the software components of the devices on the field is one of the most requested functionalities by the Telit OneEdge users.

An IoT device can have different software components, listed below:

- a. Firmware of the Telit modem, always present;
- b. AppZone application, only in hostless architecture;
- c. External MCU image, only in hosted architecture;
- d. Configuration files, for example custom LwM2M objects, security certificates, settings for the host application;

More details about the high-level architecture are described in [chapter 3](#).

The “Firmware Over The Air” feature (point a in the list above), as known as FOTA, allows to remotely upgrade the firmware of the Telit modem on the field (but not the other software components, points b-d in the list above).

According to OMA specifications and based on “Firmware Update” object (object ID 5), the FOTA state machine of the firmware upgrade is managed by the **LwM2M client internally** and this means that the LwM2M client downloads the firmware delta file and install it internally, without or with minimal required action for the host application.

For this feature, a specific web application, called as “FOTA app”, is available on OneEdge IoT portal to manage the Firmware upgrade at User side; please refer to “OneEdge Firmware Management App” guide.

Unfortunately, the OMA “Firmware Update” object doesn’t cover the other cases from b. to d. of the list (AppZone package, the image of the external MCU, configuration files), so the logic of the state machine must be developed on **the host application itself**.

The logical difference between these 2 managements is showed in the Figure 1.

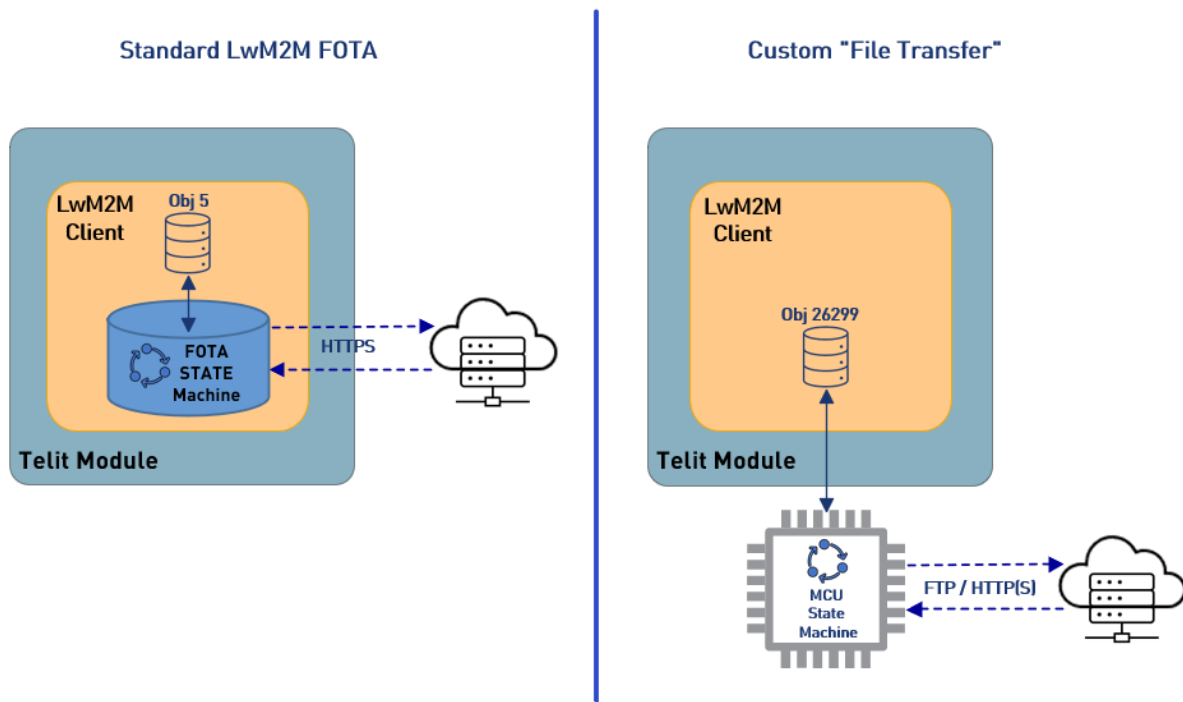


Figure 1: comparison between standard LwM2M FOTA management and custom "File Transfer" feature

Some interesting uses cases involved in generic file transfer could be:

- Force a remote device to download and install a new AppZone package;
- Force a remote device to download and apply a new image for an external host;
- Force a remote device to download a generic file (i.e. a configuration file);
- Allow a remote device to request a file to download (i.e. an expired file);
- Allow a remote device to upload a generic file (i.e. a logging file, a photo from a camera) to a specific remote location;

This application note will explain a way to perform a generic file transfer to cover most of the requested use cases.



Tip: It's highly recommended that the "OneEdge Getting started" guide is consulted and understood as precondition of this application note.

3. HIGH-LEVEL ARCHITECTURE

The main actors included in this OneEdge application are described in the below figures:

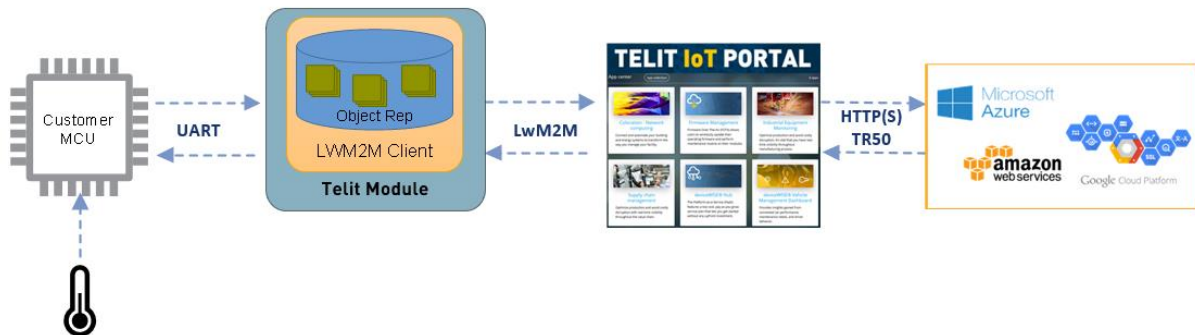


Figure 2: "hosted" high-level architecture

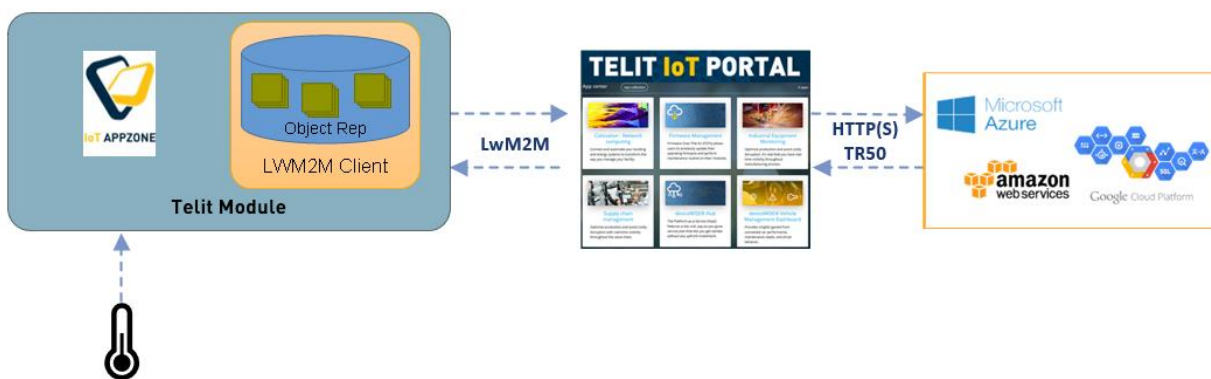


Figure 3: "hostless" high-level architecture

3.1. Customer MCU or AppZone

The "Customer MCU" is the microprocessor that manages the logic of the customer application, described in Figure 2.

Figure 3 shows an alternative architecture, in which the "Customer MCU" is replaced by an AppZone application running into the Telit Module, which performs the same actions and logics of the "Customer MCU".

Please refer to ts-emea@telit.com for more details about AppZone, which it is not in the scope of this application note.

In both cases, the application logics is in charge to the customer, based on the real use case scenario.

3.2. Telit module

The Telit module is a cellular modem with a LwM2M client built-in in the software.

The LwM2M client is responsible for communicating with the LwM2M server through the LwM2M protocol.

When customer application is running on external MCU, it communicates with the Telit modem through AT commands on the serial port.

On the other hand, when the AppZone is utilized, the customer application runs inside the Telit module and normally uses internal APIs to communicate with the main modem firmware.

3.3. Telit OneEdge IoT portal

The Telit OneEdge IoT portal is a web-based portal with “Device Management” capabilities and with an integrated LwM2M server.

Across the OneEdge service, the Telit OneEdge IoT portal is responsible for providing a high-level representation of the devices and to show their data.

Furthermore, the Telit OneEdge IoT portal can interact with external Third-party services like Microsoft Azure, Amazon AWS, Google Cloud Platform and generic HTTP/HTTPS services through connectors and TR50 interface.

3.4. External Third-party services

The Telit OneEdge IoT portal integrates some functionalities to push data to external clouds, for example Microsoft Azure or Amazon AWS.

Furthermore, TR50 interface on the Telit OneEdge IoT portal can be used to pull data or execute generic actions on the portal.

4. FILE TRANSFER IN ONEEDGE

Since LwM2M is a protocol designed for IoT applications, it is suitable and optimized for low data traffic (typically few bytes) and low frequency of communication.

The maximum size that can be transported into a unique LwM2M message is 1024 bytes through an opaque resource.

Therefore, the most convenient way to upload and download bigger application files from a remote network location is to use FTP and HTTP functionalities available in the Telit devices.

Furthermore, thanks to its data model organization, the LwM2M protocol is more suitable to manage all the data related to the transfer

The proposed approach is to use a LwM2M custom object to manage and transmit the parameters needed to perform the data transfer, that is performed via FTP/HTTP according to a state machine and rules managed by the host application.

The logical blocks involved in this process are summarized below:

- The “File Transfer Manager” **LwM2M custom object** to manage the **metadata** of the transfer;
- The real file transfer performed via **FTP/HTTP protocol**;
- The **state machine** managed by the host application.

All these blocks are described on the following paragraphs.

4.1. “File Transfer Manager” LwM2M custom object

The table below shows the structure of the “File Transfer Manager” custom object that will be used in this application note:

Res ID	Name	Description	Operations	Mandatory	Type
0	File Transfer Type	Type of transfer: 0=FILE_TRANSFER_TYPE_FILE_DOWNLOAD, 1=FILE_TRANSFER_TYPE_FILE_UPLOAD	RW	Yes	Integer
1	Filename	Name of the file being transferred	RW	Yes	String
2	File version	Version of the file	RW	No	String
3	File type	Scope of the file: i.e. "XML", "AppZone app", "MCU binary", "log", "configuration", etc.	RW	No	String
11	Transfer URL	URL path to the file. for example: ftp://server.com/test.txt; http://hosting.com/XML/object_xxx.xml; https://hosting.com/MCU/MCU_binary_v2.0	RW	Yes	String

Res ID	Name	Description	Operations	Mandatory	Type
12	Transfer parameters	Multiple instance resource to represent additional parameters for the transfer, for example: instance 0: "username=user1", instance 1: "password=qwerty"	RW	No	String
21	State	0= IDLE, 1=FILE_TRANSFER_IN_PROGRESS, 2=FILE_TRANSFER_COMPLETED, 3=FILE_TRANSFER_FAILED	R	Yes	Integer
22	Additional message	Any message related to the last transfer	R	No	String
31	Transfer URL request	This resource can be used by the host application to request a Transfer URL. It can be an integer, a timestamp, a session ID, etc. for example, the MCU wants to upload a log or wants to receive a configuration file	R	No	Integer

Table 3: "File Transfer Manager" LwM2M custom object structure

The object is defined as "multiple instance", such that the host application can map different files/application components in different instances of the "File Transfer Manager" object itself.

An example will be described in the next chapters related to use cases.



Note: the "File Transfer Manager" custom object proposed in this application note is just an example: it can be adapted removing or adding resources according to the application requirements.

4.2. FTP and HTTP management on Telit modem

The FTP and HTTP functionalities are available on Telit modem as default and these application protocols are the best way to download/upload a file from an external server.

Both the protocols need a PDP context activated.

The following command must be performed by the host application to open the context identifier:

```
AT#SGACT=n,1
#SGACT: xxx.xxx.xxx.xxx
OK
```

When successful, the command returns the IP address of the application session.

Another precondition related to these application protocols is to enable the HW flow control on both the sides (host application and modem) of the serial port.

- On modem side the HW flow control must be enabled through the AT&K3 command.
- The specific setting to enable it on the host application side depends on the external MCU used.

In case of TATC the HW flow control setting is showed in Figure 4:

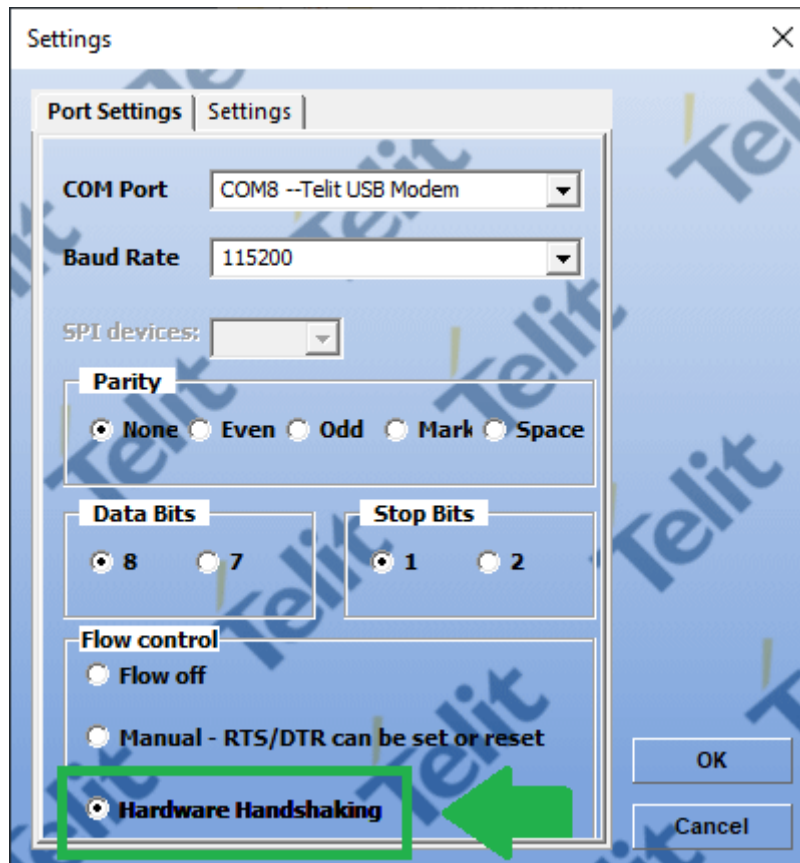


Figure 4: Flow control set as “Hardware Handshaking” on TATC

4.2.1. Basic AT commands for FTP

The logical steps to download/upload a file from an FTP server are three:

- a. the establishment of FTP connection
- b. the folder selection on the server
- c. the file get or put action

The basic AT commands for FTP are briefly described below, but please refer to the “AT command reference guide” for a complete description and “Telit IP Easy User Guide” for examples.

4.2.1.1. AT#FTPCFG

This command is used to customize basic FTP parameters like the timeout of the connection.

The syntax is:

```
AT#FTPCFG=<tout>,<IPPignoring>[,<FTPSEn>[,<FTPext>]]
```

4.2.1.2. AT#FTPOPEN

This command is used to open an FTP connection to a server.

Most important parameters are the server address and port,

The syntax is:

```
AT#FTPOPEN=[<server:port>,<username>,<password>,<mode>[,<cid>]]
```

The <server:port> parameter is the most important one.

The <mode> parameter is useful to select between active and passive mode; the default value is 0 as “active”

4.2.1.3. AT#FTPCWD

This command is used to change the working directory on the FTP server.

The syntax is:

```
AT#FTPCWD=[<dirname>]
```

4.2.1.4. AT#FTPFSIZE

This command is useful to retrieve the size of a file in bytes.

The syntax is:

```
AT#FTPFSIZE=<filename>
```

4.2.1.5. AT#FTPGET

This command performs a download of the content of the file from the FTP server.

The syntax is:

```
AT#FTPGET=<filename>
```

4.2.1.6. AT#FTPPUT

This command performs an upload of data to a file on the FTP server.

The syntax is:

```
AT#FTPPUT=<filename>[,<connMode>]
```

The <connMode> parameter is used to select the way to provide the data to the modem interface: in “online mode” (value 0, as default), the data is injected after the CONNECT message.

Otherwise, in “command mode” (value 1), the AT#FTPAPPEXT command must be used to inject the data to the file.

4.2.1.7. Example

Suppose that the host application has to access the file “test.txt” from the server path ftp:// server/folder/, the most important parameters to be identified are:

- Server address=“server”
- Filename=“test.txt”
- Folder=“folder”
- authentication through
 - Username=“user123”
 - Password=“qwerty”

The procedure is the following:

a. Connect to the FTP server

```
AT#FTPOPEN="server","user123","qwerty",1  
OK
```

b. Select the right folder, if needed

```
AT#FTPCWD="folder"  
OK
```

c. Retrieve the file size of the file

```
AT#FTPFSIZE="test.txt"  
#FTPFSIZE: 6  
OK
```

d. Get the content of the file

```
AT#FTPGET="test.txt"  
CONNECT
```


abcdef
NO CARRIER

- e. The host application must read the file content (in this example “**abcdef**”) from the serial port and manage it according to the specific scope.



Tip: the command `AT#FTPLIST` could be useful to get the size of the file before to download it.



Tip: the command `AT#FTPPUT` must be used at step c to upload a file on the current folder of the FTP server

4.2.1.8. FTPS

The secured extension of FTP, as known as FTPS, FTPES, FTP-SSL or FTP Secure, is disabled by default but it can be enabled through the parameter `FTPSEn` of `AT#FTPCFG` command:

```
AT#FTPCFG=<tout>,<IPPIgnoring>[,<FTPSEn>[,<FTPext>]]
```

<FTPSEn> values:

0 : disable FTPS security (default)

1 : enable FTPS security

4.2.2. Basic AT commands for HTTP

The logical steps to download/upload a file from a HTTP server are similar to FTP case.

The basic AT commands for HTTP are briefly described below, but please refer to the “AT command reference guide” for a complete description and “Telit IP Easy User Guide” for examples.

4.2.2.1. AT#HTTPCFG

This command is the most important for HTTP connection because is used to customize main parameters of the connection, like the server address and the type of authentication

The syntax is:



```
AT#HTTPCFG=<prof_id>[,<server_address>[,<server_port>[,<auth_type>[,<username>[,<password>[,<ssl_enabled>[,<timeout>[,<cid>[,<pkt_size>[,<unused1>[,<unused2>]]]]]]]]]]]
```

<auth_type> values:

0 : no authentication (default)

1 : basic authentication, through username and password

4.2.2.2. AT#HTTPQRY

This command is used to perform GET, HEAD and DELETE actions to an HTTP server through the **<command>** parameter.

The syntax is:

```
AT#HTTPQRY=<prof_id>,<command>,<resource>[,<extra_header_line>]
```

<command> values:

0 : GET

1 : HEAD

2: DELETE

The response of the server is showed on the following URC message:

```
#HTTPRING:<prof_id>,<http_status_code>,<content_type>,<data_size>
```

4.2.2.3. AT#HTTPRCV

This command is used to read data from HTTP server.

The syntax is:

```
AT#HTTPRCV=<profId>[,<maxByte>]
```

4.2.2.4. AT#HTTPSND

This command is used to perform POST and PUT actions to an HTTP server through the **<command>** parameter.

```
AT#HTTPSND=<profId>,<command>,<resource>,<dataLen>[,<postParam>[,<extraHeaderLine>]]
```

<command> values:

0 : POST

1 : PUT

4.2.2.5. Example

Suppose that the host application has to access the file on the path "http://**server**/**folder**/**test.txt**", the following parameter must be identified:

- Server address="**server**"
- No HTTPS, so no SSL encryption and authentication
- Filename="**test.txt**"
- Folder="**folder**"
- Basic authentication through
 - Username="**user123**"
 - Password="**qwerty**"

- a. Configure the connection to the HTTP server

```
AT#HTTPCFG=0,"server",80,1,"user123","qwerty",0
OK
```

- b. Only in case of download, query the file and get the type and size ("text/plain" and 61440 bytes in this example)

```
AT#HTTPQRY=0,0,"/folder/test.bin"
OK
#HTTPRING: 0,200,"text/plain",61440
```

- c. In case of **download**, get the content of the file chunk by chunk of 1024 bytes:

```
AT#HTTPRCV=0,1024
<<<  first block of data
OK
AT#HTTPRCV=0,1024
<<<  second block of data
OK
...
AT#HTTPRCV=0,1024
<<<  last block of data
OK
```

- d. In case of **upload**, the command AT#HTTPSND must be used, providing the size (in this example 61440 bytes) and the extension (1 for "text/plain" HTTP content type)

```
AT#HTTPSND=0,0,"/folder/test.txt",61440,1
binary data here
OK
```

- e. The host application must read the data content from the serial port and manage it according to the specific scope.

4.2.2.6. HTTP vs HTTPS

The example in [paragraph 4.2.2.5](#) is based on HTTP communication without any TLS/SSL authentication and encryption.

The HTTPS protocol can be used to add more security to the file transfer.



Warning: In order to use SSL on HTTP, the “secure SSL socket” feature (AT#SSLEN) must be kept disabled (as default) on the modem:

```
AT#SSLEN?  
#SSLEN: 1,0  
OK
```

4.2.2.6.1. SSL encryption

The `<ssl_enabled>` parameter of AT#HTTPCFG command can be set to 1 to enable the SSL encryption:

```
AT#HTTPCFG=<prof_id>[,<server_address>[,<server_port>[,<auth_type>[,<u  
sername>[,<password>[,<ssl_enabled>[,<timeout>[,<cid>[,<pkt_size>[,<un  
used1>[,<unused2>]]]]]]]]]]]
```

<ssl_enabled> values:

0 : SSL encryption disabled (default)

1 : SSL encryption enabled

4.2.2.6.2. SSL authentication

If SSL encryption is enabled on AT#HTTPCFG (ssl_enable=1), additional SSL security parameters for HTTPS can be configured through AT#SSLSECCFG command; in this command, the most important parameter is the third, related to the SSL authentication mode:

```
AT#SSLSECCFG=<SSId>,<CipherSuite>,<auth_mode>
```

<auth_mode> values:

0 : SSL Verify None (default)

1 : manage server authentication

2 : manage server and client authentication if requested by the remote server

Summary of possible configurations related to HTTP and HTTPS:

- HTTP with “no username and password authentication”, no SSL
AT#HTTPCFG=0,"server",80,0
- HTTP with “basic authentication” (username and password), no SSL encryption
AT#HTTPCFG=0,"server",80,1,"user123","qwerty",0
- HTTPS with SSL encryption but no SSL authentication:
AT#SSLSECCFG=1,0,0
AT#HTTPCFG=0,"server",443,0,"","",1
- HTTPS with SSL encryption and SSL authentication (both client and server):
AT#SSLSECCFG=1,0,2
AT#SSLSECDATA=1,1,0,<certificate_size>
(inject pem certificate here)
AT#HTTPCFG=0,"server",443,0,"","",1



Warning: Refer to “AT command guide” and “Telit IP Easy User Guide” for more details about the AT commands

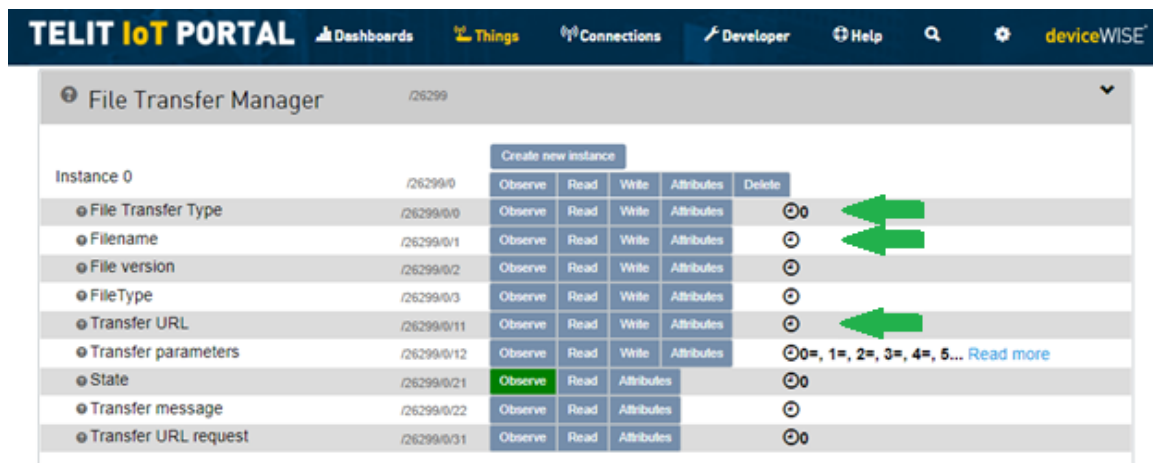
4.3. State machine and rules

Since this procedure doesn't follow any OMA specification, the developer of the host application could define its own state machine on host application side and rules shared with the Portal User on server side to put together the LwM2M object and the real transfer.

The rules applied in this application note are:

- On server side, the **Portal User** must be able to **initiate a file transfer**, writing the following parameters to LwM2M resources of “File Transfer Manager” object:
 - the transfer type (download or upload) on resource 0,
 - the filename on resource 1,
 - the transfer URL on resource 11,
 - and, optionally:
 - additional transfer parameters on resource 12,

- o the file type on resource 3



TELIT IoT PORTAL					
File Transfer Manager /26299					
Create new instance					
Instance 0	/26299/0	Observe	Read	Write	Delete
File Transfer Type	/26299/0/0	Observe	Read	Write	Attributes
Filename	/26299/0/1	Observe	Read	Write	Attributes
File version	/26299/0/2	Observe	Read	Write	Attributes
FileType	/26299/0/3	Observe	Read	Write	Attributes
Transfer URL	/26299/0/11	Observe	Read	Write	Attributes
Transfer parameters	/26299/0/12	Observe	Read	Write	Attributes
State	/26299/0/21	Observe	Read	Attributes	
Transfer message	/26299/0/22	Observe	Read	Attributes	
Transfer URL request	/26299/0/31	Observe	Read	Attributes	

Figure 5: resources of the object to be written

- b. When the host application detects a **new value on “Transfer URL” (resource 11)**, it must read all the other values of the object and perform the FTP/HTTP transfer according to the provided parameters (refer to [paragraph 4.2](#));



Tip: the AT#LWM2MMON is useful to let the host application to receive URC messages when the resources are modified

- c. The host application changes the “State” resource value according to the progress of the file transfer. Furthermore, by setting an observation on this resource, the LwM2M client will notify the state change to server:
- when the host application starts an FTP/HTTP connection, it modifies the value from “idle” to “**file transfer in progress**”;
 - when the transfer is completed successfully, the host application must change value to “**file transfer completed**”;
 - In case of failure during the FTP/HTTP transfer, the host application must write the “**file transfer failed**” value to the LwM2M “state” resource;
 - Every time the host application receives a new file transfer URL, the “State” resource must be set back to “idle”;
- d. The developer must define a method to retrieve the file version and apply the value to the “**File version**” resource
- e. Optionally more information about the transfer result can be added on “**transfer message**” resource



Note: on device side, the host application can initiate a file transfer on-demand, changing the value of “**transfer URL request**” resource: in this case, the User on server side should detect this event and provide a transfer URL according to the same steps of “initiated by User”. For example, it’s the case that the host application wants to download a new certificate or needs a dynamic link to upload a log file.

4.4. Data management at host application level

Related to a downloaded file, once the host application has received a blob of data through the FTP and HTTP protocols, it must be able to handle it and apply the proper actions, for example storing it temporarily in a buffer.

The host application could, for example, use the “File type” resource to apply the appropriate actions.

Based on the file type and contents, then the received blob of data must be consumed by the application host. Below are some examples:

- a value like “XML” or “AppZone app” could mean that the file must be saved into the **modem file system** (respectively “/XML” and “/mod” folders);
- in case of “MCU binary” or “configuration”, the downloaded file will be used to upgrade the **external MCU**: the process is MCU specific and implementation is left to the developer.

For example, it could be performed by writing the data blob into specific registers or dedicated space in the memory of the MCU;

- another key word could be “log” to manage the data collected by the host application to be uploaded on a remote server for further analysis.

In case of upload, the MCU could temporarily store the blob of data in its own buffering system and then push it to FTP/HTTP data link.

4.4.1. Basic AT commands for modem File System management

The basic AT commands to manage the file system of the modem are briefly described below, but please refer to the “AT command reference guide” for a complete description.

4.4.1.1. AT#M2MLIST

The command AT#M2MLIST is used to list the content of a directory in the modem File System. The syntax is

```
AT#M2MLIST[=<path>]
```

Example to list the content of “/XML” directory:

```
AT#M2MLIST="/XML"
#M2MLIST: <.>
#M2MLIST: <..>
#M2MLIST: "connagent.xml",15664
#M2MLIST: "empty",1
#M2MLIST: "object_26299.xml",6481
#M2MLIST: free bytes: 3252224
```

4.4.1.2. AT#M2MWRITE

The AT command to write data into a file is AT#M2MWRITE. The syntax is:

```
AT#M2MWRITE=<file_name>,<size>[,<binToMod>]
```

<file_name> is a string and can be

- full file name path starting with “/”
- relative file name path
- file name (in current directory)

<size> is the size in bytes of the data to be written

<binToMod> is used to force the file to be written in the “/mod” directory



Note: the AT#M2MWRITE overwrites the content if the file already exists, otherwise it creates a new file.

Example:

The host application received the data content “**abcdef**” from FTP and write it to the file “incoming_data.txt” the “test” directory:

```
AT#FTPGET="test.txt"
CONNECT
```




```
abcdef
NO CARRIER
AT#M2MWRITE="/test/incoming_data.txt",6
>>>
abcdef
OK
```

4.4.1.3. AT#M2MREAD

The command AT#M2MREAD is used to read the content of a file stored into the modem file system. The syntax is:

```
AT#M2MREAD=<file_name>
```

<file_name> has the same rules of AT#M2MWRITE command

Example to read the content of LwM2M custom object stored in the modem file system:

```
AT#M2MREAD="/XML/object_26299.xml"
<<<<?xml version="1.0" encoding="UTF-8"?>
<!--
Copyright (c) 2020 Telit
THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, ...
-->
<LWM2M ... >
    <Object ObjectType="M0Definition">
        <Name>File Transfer Manager</Name>
        ...
        ...
    </Resources>
    <Description2/>
</Object>
</LWM2M>
OK
```

5. EXAMPLES

In this chapter two different use cases will be described:

- Use case 1: “**File Transfer triggered by Server**”: in this example (paragraph 5.3), the Server (Portal User, external application on portal side or OneEdge campaign) requests the host application to download data from an HTTPS path and replace an existing file on the modem file system
- Use case 2: “**File Transfer triggered by Host application**”: in this example (paragraph 5.4), the host application asks for the server to provide a link to upload a file.

5.1. Data flow diagram

The image below shows the flow of AT commands, LwM2M messages and events across the actors involved in the File Transfer use case.

The vertical arrows indicate the time, starting from the top to the bottom.

The first block is related to the LwM2M registration and it's common to all the use cases; the other two blocks are related to the use cases described in this chapter.

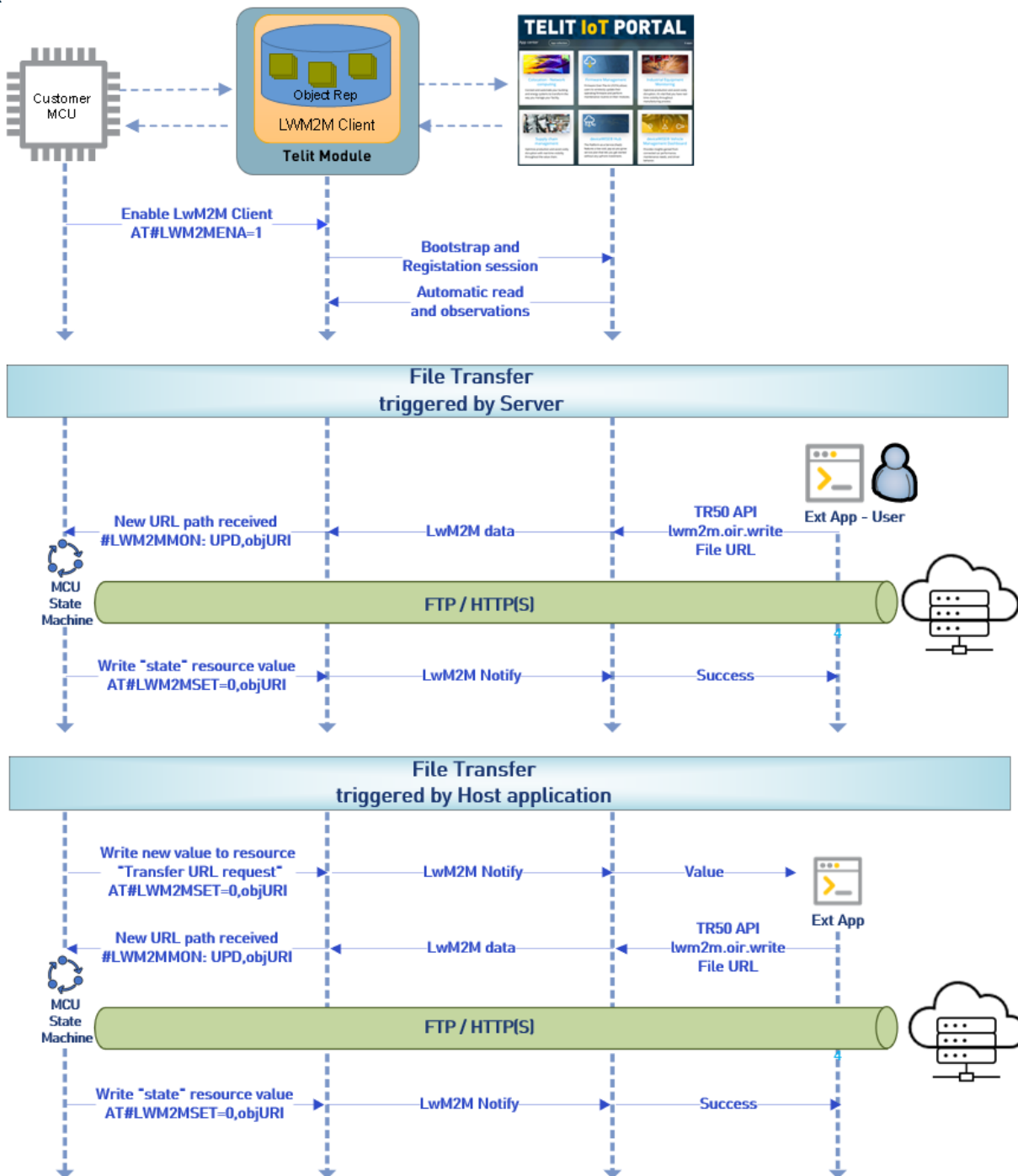


Figure 6: diagram of main actions involved

5.2. Pre-requisites and one-time configuration:

The following steps are needed as precondition.

- Materials (device, cables, SIM card, etc) and setup according to "OneEdge Getting started" document;

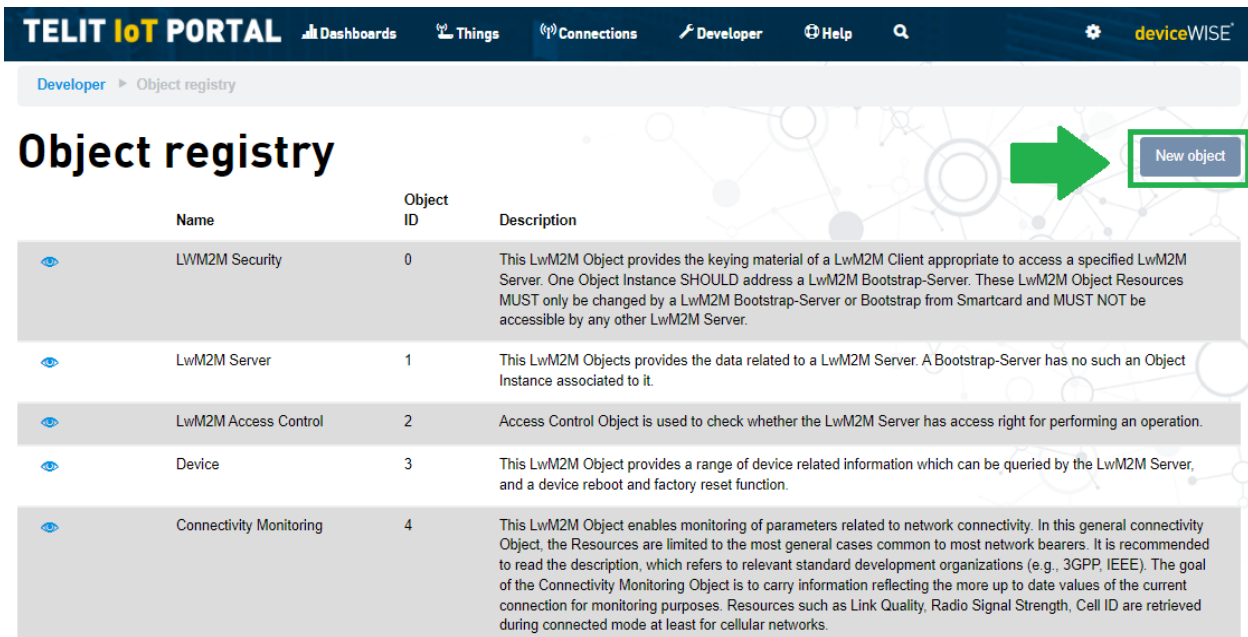
- b) download the XML file from the following link:

https://github.com/telit/oneedge-projects-resources/tree/main/use-cases/file-transfer-manager/XML/object_26299.xml

- c) Custom object file: add the file “object_26299.xml” on section “Developer → Object registry” of OneEdge IoT portal;

the 2 steps below must be performed:

- Press “New object” button to open the edit box popup
- Copy the content of the XML file and paste it on the edit popup



TELIT IoT PORTAL Dashboards Things Connections Developer Help deviceWISE

Developer ▸ Object registry

Object registry

	Name	Object ID	Description
	LWM2M Security	0	This LwM2M Object provides the keying material of a LwM2M Client appropriate to access a specified LwM2M Server. One Object Instance SHOULD address a LwM2M Bootstrap-Server. These LwM2M Object Resources MUST only be changed by a LwM2M Bootstrap-Server or Bootstrap from Smartcard and MUST NOT be accessible by any other LwM2M Server.
	LwM2M Server	1	This LwM2M Objects provides the data related to a LwM2M Server. A Bootstrap-Server has no such an Object Instance associated to it.
	LwM2M Access Control	2	Access Control Object is used to check whether the LwM2M Server has access right for performing an operation.
	Device	3	This LwM2M Object provides a range of device related information which can be queried by the LwM2M Server, and a device reboot and factory reset function.
	Connectivity Monitoring	4	This LwM2M Object enables monitoring of parameters related to network connectivity. In this general connectivity Object, the Resources are limited to the most general cases common to most network bearers. It is recommended to read the description, which refers to relevant standard development organizations (e.g., 3GPP, IEEE). The goal of the Connectivity Monitoring Object is to carry information reflecting the more up to date values of the current connection for monitoring purposes. Resources such as Link Quality, Radio Signal Strength, Cell ID are retrieved during connected mode at least for cellular networks.

New object

Figure 7: “Object registry” section with “Import” button in detail

- d) Load XML files into the device:

```
AT#M2MWRITE="/XML/object_26299.xml",6481
>>>
OK
```

- e) Reset the LwM2M client and reboot the device

```
AT#LWM2MSTS=0,999,"coaps://bs.telit.io",1
OK
AT#REBOOT
OK
```

- f) Enable the LwM2M client and wait for registration to the IoT portal

```
AT#LWM2MW=0,33211,0,0,1,1
OK
```

```
AT#LWM2MENA=1
OK
LWM2M-TLT:"BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"REGISTERING",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT:"REGISTERED",SSID=99,"coaps://api-dev.devicewise.com"

AT#LWM2MW=0,33211,0,0,1,1
OK
```

5.3. Use case 1: File Transfer triggered by Server

This use case is related to the Server (a Portal User, an external application on portal side or a OneEdge campaign) that ask for the host application to download or upload a file.

In particular, the example in this paragraph is related to the download of a data content from an HTTPS storage and replace an existing file, a Lwm2M custom object file, in the modem file system with a new version of the object.

The same approach can be used to upgrade a new AppZone package or a new SSL certificate.

The step-by-step procedure is reported below:

1. If it has not already been done, start the Lwm2M client

```
AT#LWM2MENA=1
OK
LWM2M-TLT:"REGISTERING",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT:"REGISTERED",SSID=99,"coaps://api-dev.devicewise.com"
```

2. Create an instance of the custom object

```
AT#LWM2MNEWINST=0,26299,0
OK
```

Tip: The host application could map all the SW components (for example custom LwM2M objects, MCU image, AppZone binary) into the instances of the “File Transfer Manager” objects that are configured during end product assembly process.

For example:



AT#LWM2MW=0,26299,0,1,0,"object_26299"

AT#LWM2MW=0,26299,0,2,0,"1.0"

AT#LWM2MW=0,26299,0,3,0,"XML LwM2M"

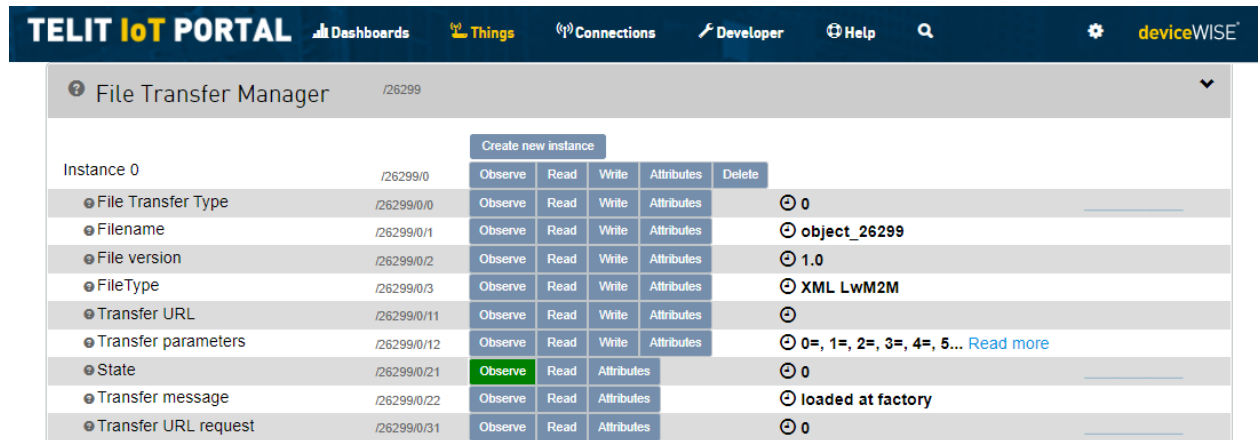
AT#LWM2MSET=2,26299,0,22,0,"loaded at factory"

In this way a Portal User on server side (or an external application or a OneEdge campaign) is able to read which SW components and versions are present on each remote device.

TELIT IoT PORTAL						
Dashboards Things Connections Developer Help deviceWISE						
File Transfer Manager /26299						
Create new instance						
Instance 0	/26299/0	Observe	Read	Write	Attributes	Delete
File Transfer Type	/26299/0/0	Observe	Read	Write	Attributes	0
Filename	/26299/0/1	Observe	Read	Write	Attributes	object_26299
File version	/26299/0/2	Observe	Read	Write	Attributes	1.0
FileType	/26299/0/3	Observe	Read	Write	Attributes	XML LwM2M
Transfer URL	/26299/0/11	Observe	Read	Write	Attributes	
Transfer parameters	/26299/0/12	Observe	Read	Write	Attributes	0=, 1=, 2=, 3=, 4=, 5... Read more
State	/26299/0/21	Observe	Read	Attributes		0
Transfer message	/26299/0/22	Observe	Read	Attributes		loaded at factory
Transfer URL request	/26299/0/31	Observe	Read	Attributes		0
Instance 1	/26299/1	Observe	Read	Write	Attributes	Delete
File Transfer Type	/26299/1/0	Observe	Read	Write	Attributes	0
Filename	/26299/1/1	Observe	Read	Write	Attributes	MCU_image
File version	/26299/1/2	Observe	Read	Write	Attributes	1.0.3
FileType	/26299/1/3	Observe	Read	Write	Attributes	MCU binary
Transfer URL	/26299/1/11	Observe	Read	Write	Attributes	
Transfer parameters	/26299/1/12	Observe	Read	Write	Attributes	0=, 1=, 2=, 3=, 4=, 5... Read more
State	/26299/1/21	Observe	Read	Attributes		0
Transfer message	/26299/1/22	Observe	Read	Attributes		loaded at factory
Transfer URL request	/26299/1/31	Observe	Read	Attributes		0
Instance 2	/26299/2	Observe	Read	Write	Attributes	Delete
File Transfer Type	/26299/2/0	Observe	Read	Write	Attributes	0
Filename	/26299/2/1	Observe	Read	Write	Attributes	m2mapz.bin
File version	/26299/2/2	Observe	Read	Write	Attributes	1.0
FileType	/26299/2/3	Observe	Read	Write	Attributes	appzone app
Transfer URL	/26299/2/11	Observe	Read	Write	Attributes	
Transfer parameters	/26299/2/12	Observe	Read	Write	Attributes	0=, 1=, 2=, 3=, 4=, 5... Read more
State	/26299/2/21	Observe	Read	Attributes		0
Transfer message	/26299/2/22	Observe	Read	Attributes		loaded at factory
Transfer URL request	/26299/2/31	Observe	Read	Attributes		0

Figure 8: example of 3 instances with 3 SW components loaded at factory

- From server side, activate an observation on the “State” resource (URI 26299/0/21) of the “File Transfer Manager” object



The screenshot shows the Telit IoT Portal interface. The main header includes 'TELIT IoT PORTAL' and navigation links for Dashboards, Things, Connections, Developer, Help, and a search icon. The 'deviceWISE' logo is in the top right. The main content area is titled 'File Transfer Manager' with a sub-header '/26299'. Below this, there's a 'Create new instance' button and a table for Instance 0. The table lists various resources and their observation status. The 'State' resource (URI /26299/0/21) is highlighted with a green 'Observe' button, indicating an active observation.

Resource	URI	Observe	Read	Write	Attributes	Delete	Value
File Transfer Type	/26299/0/0	Observe	Read	Write	Attributes		0
Filename	/26299/0/1	Observe	Read	Write	Attributes		object_26299
File version	/26299/0/2	Observe	Read	Write	Attributes		1.0
FileType	/26299/0/3	Observe	Read	Write	Attributes		XML LwM2M
Transfer URL	/26299/0/11	Observe	Read	Write	Attributes		
Transfer parameters	/26299/0/12	Observe	Read	Write	Attributes		0=, 1=, 2=, 3=, 4=, 5... Read more
State	/26299/0/21	Observe	Read	Attributes			0
Transfer message	/26299/0/22	Observe	Read	Attributes			loaded at factory
Transfer URL request	/26299/0/31	Observe	Read	Attributes			0

Figure 9: observation active on instance 0

- On modem side, configure ACK Notify service on the LwM2M client to receive URC on host application when the notifies are received by the OneEdge IoT portal

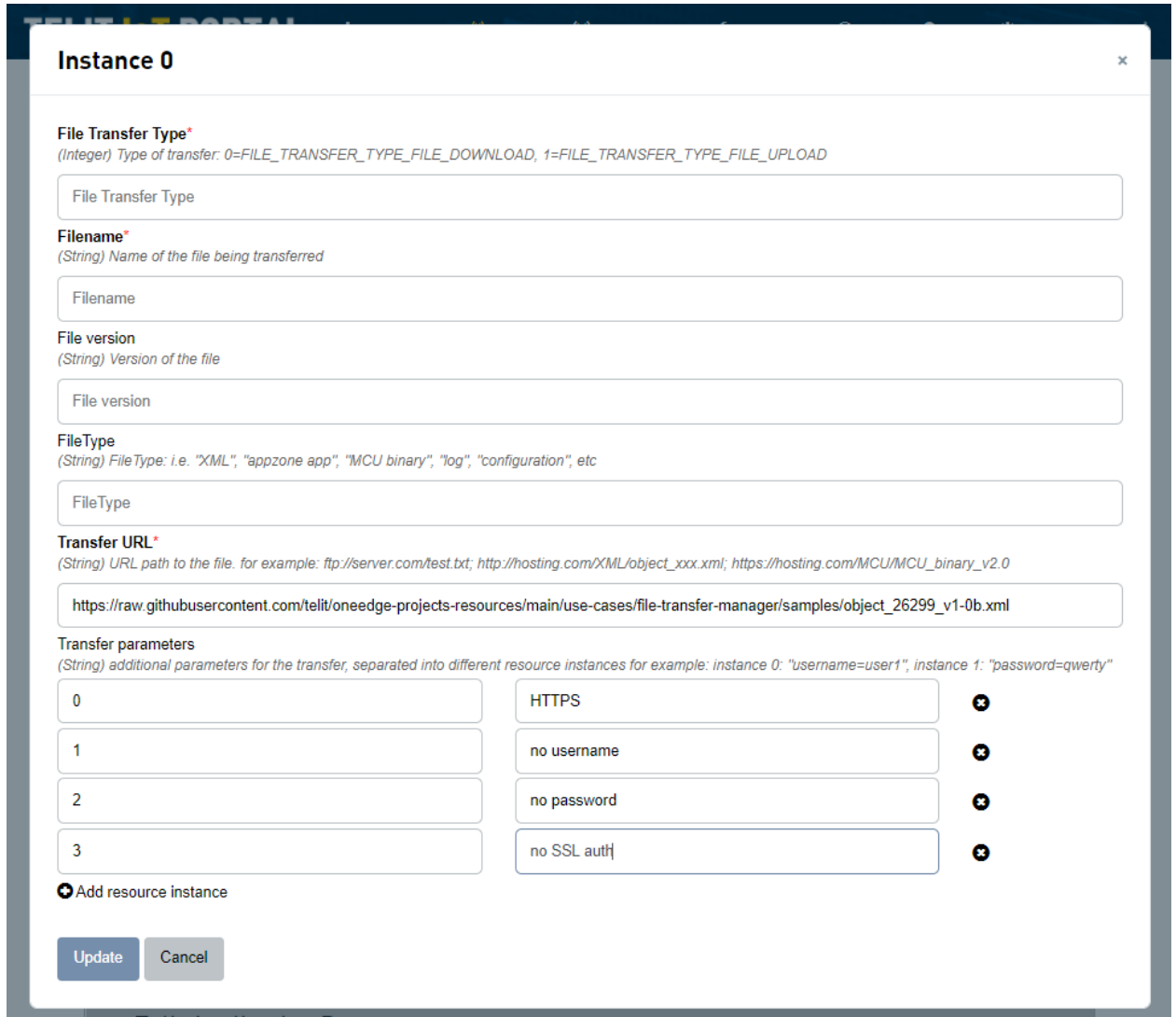
```
AT#LWM2MNFYACKURI=0,1,26299,0,21
OK
AT#LWM2MNFYACKENA=0,1
OK
```

- On modem side, configure monitoring service on the LwM2M client to receive the URC messages on host application when a remote setting is received on the LwM2M client

```
AT#LWM2MMON=1,26299
OK
```

- On server side, write the following values on the instance 0 of the “File Transfer Manager” object:

- Resource 11 = https://raw.githubusercontent.com/telit/oneedge-projects-resources/main/use-cases/file-transfer-manager/samples/object_26299_v1-0b.xml
- Resource 12, free parameters to customize the file transfer, for example: {"0"="HTTPS", "1"="no username", "2"="no password", "3"="no SSL auth"}



Instance 0

File Transfer Type*
(Integer) Type of transfer: 0=FILE_TRANSFER_TYPE_FILE_DOWNLOAD, 1=FILE_TRANSFER_TYPE_FILE_UPLOAD

File Transfer Type

Filename*
(String) Name of the file being transferred

Filename

File version
(String) Version of the file

File version

FileType
(String) FileType: i.e. "XML", "appzone app", "MCU binary", "log", "configuration", etc

FileType

Transfer URL*
(String) URL path to the file. for example: ftp://server.com/test.txt; http://hosting.com/XML/object_XXX.xml; https://hosting.com/MCU/MCU_binary_v2.0

https://raw.githubusercontent.com/telit/oneedge-projects-resources/main/use-cases/file-transfer-manager/samples/object_26299_v1-0b.xml

Transfer parameters
(String) additional parameters for the transfer, separated into different resource instances for example: instance 0: "username=user1", instance 1: "password=querty"

0	HTTPS	✖
1	no username	✖
2	no password	✖
3	no SSL auth	✖

➕ Add resource instance

Update Cancel

Figure 10: writing multiple resources of instance 0 on OneEdge IoT portal

The same “write” operation can be done through the TR50 interface of OneEdge IoT portal:

```
{
  "data": {
    "command": "lwm2m.device.oir.write",
    "thingId": <thingId>,
    "objId": 26299,
    "instId": 0,
    "encoding": 11542,
```



```

    "params": {
        "data": {
            "11": "https:\\\\raw.githubusercontent.com\\telit\\oneedge-projects-resources\\main\\use-cases\\file-transfer-manager\\samples\\object_26299_v1-0b.xml",
            "12": {"0":"HTTPS", "1":"no username", "2":"no password", "3":"no SSL auth"}
        }
    }
}

```

OneEdge Tip: the above action could be performed through a campaign on OneEdge IoT portal in order to send the same transfer path to multiple devices at the same time.



Furthermore, the “File version” resource can be read by the campaign to send the new file transfer parameters only to devices that match a specific file version.

For an example of a campaign of OneEdge IoT portal, refer to paragraph 5.3.1.

7. On modem side, the host application will receive the following URC messages on the serial port:

```

#LWM2MMON: UPD, "/26299/0/11/0"
#LWM2MMON: UPD, "/26299/0/12/0"
#LWM2MMON: UPD, "/26299/0/12/1"
#LWM2MMON: UPD, "/26299/0/12/2"
#LWM2MMON: UPD, "/26299/0/12/3"

```

8. The host application reads the values of the resources:

```

AT#LWM2MR=0,26299,0,0,0
#LWM2MR: 0
OK
AT#LWM2MR=0,26299,0,1,0
#LWM2MR: "object_26299"
OK

```

```
AT#LWM2MR=0,26299,0,3,0
#LWM2MR: "XML LwM2M"
OK
AT#LWM2MR=0,26299,0,11,0
#LWM2MR: "https://raw.githubusercontent.com/telit/oneedge-
projects-resources/tree/main/use-cases/file-transfer-
manager/samples/object_26299_v1-0b.xml"
OK
AT#LWM2MR=0,26299,0,12,0
#LWM2MR: "HTTPS"
OK
AT#LWM2MR=0,26299,0,12,1
#LWM2MR: "no username"
OK
AT#LWM2MR=0,26299,0,12,2
#LWM2MR: "no password"
OK
AT#LWM2MR=0,26299,0,12,3
#LWM2MR: "no SSL auth"
OK
```

Tip: The resources of the same instances can be read all together also through the AT#LWM2MOBJGET command:

```
AT#LWM2MOBJGET=0,26299,0
{"bn":"/26299/0/",
"e":[
{"n":"0","v":0},
{"n":"1","sv":"object_26299"},
{"n":"2","sv":"1.0"},
{"n":"3","sv":"XML LwM2M"},
{"n":"11","sv":"https:\\\\raw.githubusercontent.com\\
telit\\oneedge-projects-resources\\tree\\main\\use-
cases\\file-transfer-manager\\samples\\
object_26299_v1-0b.xml"},
{"n":"12/0","sv":"HTTPS"},
{"n":"12/1","sv":"no username"},
{"n":"12/2","sv":"no password"},
{"n":"12/3","sv":"no SSL auth"},
...
{"n":"12/8","sv":""},
{"n":"12/9","sv":""},
{"n":"21","v":0},
{"n":"22","sv":"loaded at factory"},
{"n":"31","v":0}]
}
OK
```



Warning: the AT commands **AT#LWM2MOBJGET** and **AT#LWM2MOBJSET** are available only from 30.01.xx0 - M0B.xx0005 and 37.00.xx4 - M0C.xx0003 SW versions. In older SW versions the same actions can be performed through single operations on each resource with **AT#LWM2MGET/R** and **AT#LWM2MSET/W** commands

9. The host application must perform the file transfer according to the provided parameters: in this example, it's a download (resource 0=0) from an **HTTPS** server, without username and password, with no SSL authentication:

```
AT#SGACT=1,1
#SGACT: 10.4.21.156
OK
AT#SSLSECCFG=1,0,0
OK
AT#HTTPCFG=0,raw.githubusercontent.com,443,0,"","",1
OK
AT#HTTPQRY=0,0,"/telit/oneedge-projects-resources/main/use-
cases/file-transfer-manager/samples/object_26299_v1-0b.xml"
OK
```

10. The host application waits for the "HTTPRING" URC message to know the type and **size** of the file:

```
#HTTPRING: 0,200,"text/plain; charset=utf-8",6482
```

11. The host application could write the value "1" to "State" resource to notify to the server that the transfer is in **progress**

```
AT#LWM2MSET=0,26299,0,21,0,1
OK
```

Then, the ACK will be received:

```
#LWM2MNFYACK: 0,99,"/26299/0/21", "ACK"
```

12. According to the size (6482 bytes in this example), the host application could reserve a specific buffer where to store temporarily the data downloaded.

The host application gets the data of the file chunk by chunk; the length of the chunks in this example is 1440 bytes, so the host application must repeat the **AT#HTTPRCV** command 5 times (1440x4 + 722 bytes) to download all the content of the file.

```
AT#HTTPRCV=0,1440
<<<<?xml version="1.0" encoding="UTF-8"?>
<!--
Copyright (c) 2020 Telit
THE SOFTWARE IS PROVIDED "AS IS",
...
THE SOFTWARE.
-->
```

```
<LWM2M xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="http://openmobilealliance.org/tech
/profiles/LWM2M.xsd">
  <Object ObjectType="MODefinition">
    <Name>File Transfer Manager</Name>
    <Description1><![CDATA[This custom object provides file
transfer
  capabilities]]></Description1>
    <ObjectID>26299</ObjectID>
    <ObjectURN>urn:oma:lwm2m:x:26299</ObjectURN>
    <LWM2MVersion>1.0</LWM2MVersion>
    <ObjectVersion>1.0b</ObjectVersion>
    <MultipleInstances>Multiple</MultipleInstances>
    <Mandatory>Optional</Mandatory>
    <Resources>
      ...
    </Resources>
  </Object>
</LWM2M>
```

OK

AT#HTTTPRCV=0,1440

```
<<<      <Mandatory>Mandatory</Mandatory>
          <Type>Integer</Type>
          <RangeEnumeration>0-1</RangeEnumeration>
          ...
          <Item ID="3">
            <Name>FileType</Name>
            <Operations>RW</Opera
```

OK

AT#HTTTPRCV=0,1440

```
<<< tions>
          <MultipleInstances>Single</MultipleInstances>
          ...
          <Units></Units>
```

OK

AT#HTTTPRCV=0,1440

```
<<<      <Description><![CDATA[additional parameters for
the transfer,
          separated into different resource instances
          ...
          <Item ID="31">
            <Name>Transfer URL
```

OK

AT#HTTTPRCV=0,1440

```
<<< request</Name>
          <Operations>R</Operations>
          ...
          </Resources>
          <Description2/>
        </Object>
      </LWM2M>
```

OK

13. The host application writes the value "2" to "State" resource to notify to the server that the transfer is **completed**



File Transfer on OneEdge Application Note

```
AT#LWM2MSET=0,26299,0,21,0,2
OK
```

the ACK is received:

```
#LWM2MNFYACK: 0,99,"/26299/0/21","ACK"
```

14. Since the file type of the transfer is “XML LwM2M”, the host application stores it on the modem file system:

```
AT#M2MWRITE="/XML/object_26299.xml",6482
>>>
data here
OK
```

15. The host application could write the new file version value to the resource 2

```
AT#LWM2MW=0,26299,0,2,0,"1.0b"
OK
```

16. The host application could write an additional message to the resource 22:

```
AT#LWM2MSET=2,26299,0,22,0,"XML file successfully downloaded and installed"
OK
```



Tip: At step 14, according to different file types, the host application could apply different actions.

The link to download the original “object_26299.xml” file is

https://raw.githubusercontent.com/telit/oneedge-projects-resources/tree/main/use-cases/file-transfer-manager/XML/object_26299.xml

It can be used to repeat the procedure again to restore the version 1.0 of the file.

5.3.1. Campaign on OneEdge IoT portal

The write operation on step 6 of the Use case 1 can be performed on multiple devices in one shot through a campaign on OneEdge IoT portal.

In order to evaluate it, a campaign is available on the link

https://github.com/telit/oneedge-projects-resources/tree/main/use-cases/file-transfer-manager/OE_Portal_configuration/FileTransfer_triggered_by_Server_campaign.json

The procedure to import and run the campaign is below:

- a) Download the campaign file from the link above

- b) import the file "FileTransfer_triggered_by_Server_campaign.json" on section "Developer → Campaigns" of OneEdge IoT portal;

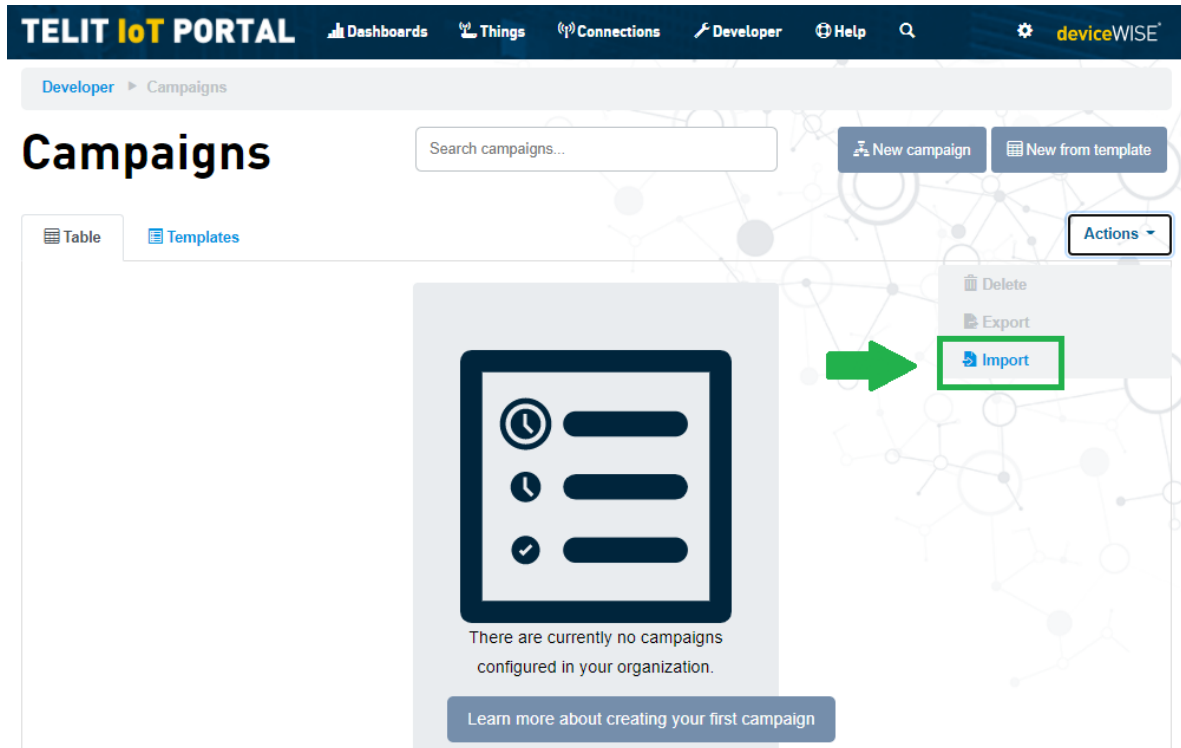


Figure 11: "Campaigns" section with "Import" button in detail

- c) Add members to the campaign, selecting the device to be applied to (All, Tags, Search, List)

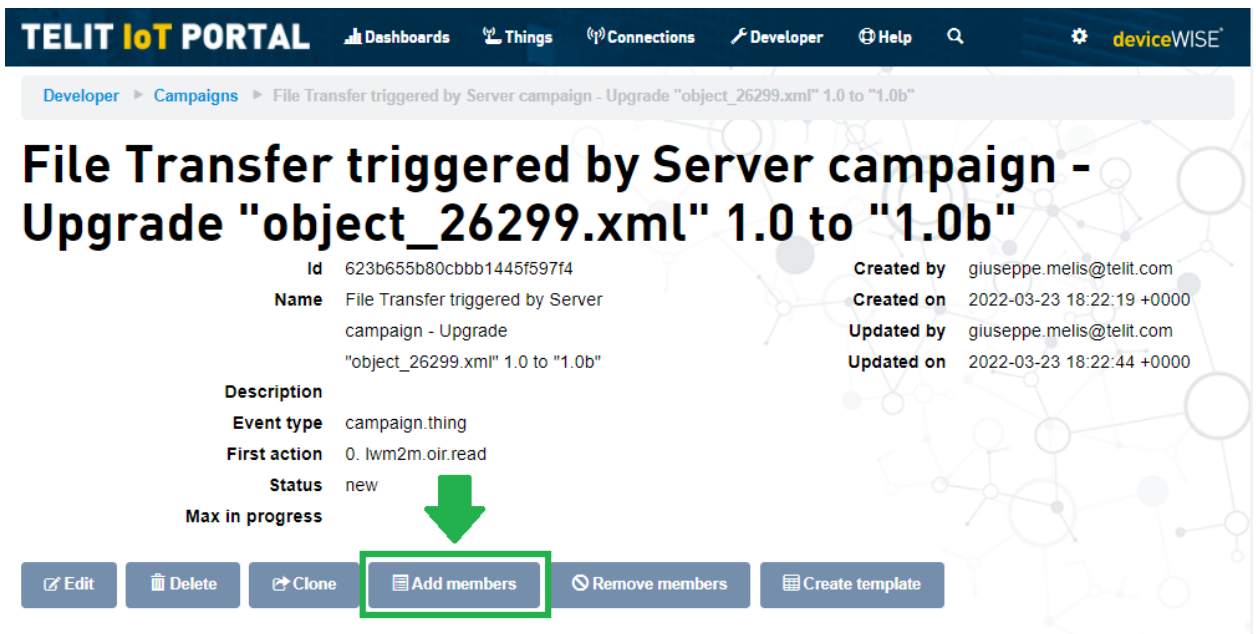


Figure 12: "Add member" button

- d) Start the campaign: all the selected devices will receive the same write operation as in step 6.

5.4. Use case 2: File Transfer triggered by Host application

This use case is related to the host application that ask for receiving a link where to upload a file or where to download it from.

This use case can be separated into 3 parts:

- a) The host application requests a file transfer URL to the server side;
- b) An "intelligent" agent on server side must generate a suitable URL depending on his rules; the "intelligent" agent could be an **external application** that manages the LwM2M server interface or a **trigger** in OneEdge environment;

Note: Below are some examples of URL that the "intelligent" agent could provide:



- a static folder in an FTP/HTTP server for all the devices
- a static folder in an FTP/HTTP server for every device
- a **dynamic** HTTP link valid for a specific time window



OneEdge Tip: a trigger on OneEdge IoT portal could be used to develop the "intelligent" agent logic

- c) The external application sends the transfer URL and other parameters to the remote device. Note that this logic block is the same as Use Case 1.

In this specific example, the host application requires a link where to upload a log file and the "intelligent" agent provides FTP parameters related to the link.

1. If it has not already been done, start the LwM2M client

```
AT#LWM2MENA=1
OK
LWM2M-TLT:"REGISTERING", SSID=99, "coaps://api-dev.devicewise.com"
LWM2M-TLT:"REGISTERED", SSID=99, "coaps://api-dev.devicewise.com"
```

2. Create an instance of the custom object

```
AT#LWM2MNEWINST=0,26299,0
OK
```

- From server side, activate an observation on the “state” (URI 26299/0/21) and on “Transfer URL request” (URI 26299/0/31) resources of the “File Transfer Manager” object

File Transfer Manager /26299		Create new instance				
Instance 0	/26299/0	Observe	Read	Write	Attributes	Delete
File Transfer Type	/26299/0/0	Observe	Read	Write	Attributes	0
Filename	/26299/0/1	Observe	Read	Write	Attributes	0
File version	/26299/0/2	Observe	Read	Write	Attributes	0
FileType	/26299/0/3	Observe	Read	Write	Attributes	0
Transfer URL	/26299/0/11	Observe	Read	Write	Attributes	0
Transfer parameters	/26299/0/12	Observe	Read	Write	Attributes	0=, 1=, 2=, 3=, 4=, 5... Read more
State	/26299/0/21	Observe	Read	Attributes		0
Transfer message	/26299/0/22	Observe	Read	Attributes		0
Transfer URL request	/26299/0/31	Observe	Read	Attributes		0

Figure 13: observations active on resources 21 and 31

- On modem side, configure ACK Notify service on the LwM2M client to receive URC on host application when the notifies are received by the OneEdge IoT portal

```
AT#LWM2MNFYACKURI=0,1,26299,0,21
OK
AT#LWM2MNFYACKURI=0,1,26299,0,31
OK
AT#LWM2MNFYACKENA=0,1
OK
```

- On modem side, configure monitoring service on the LwM2M client to receive the URC messages on host application when a remote setting is received on the LwM2M client

```
AT#LWM2MMON=1,26299
OK
```

- The host application configures the parameters for the transfer that is going to ask to the “intelligent” agent: upload as transfer type, “debug_2022-01-01.log” as filename, “log” as file type

```
AT#LWM2MW=0,26299,0,0,0,1
OK
AT#LWM2MW=0,26299,0,1,0,"debug_2022-01-01.log"
OK
AT#LWM2MW=0,26299,0,3,0,"log"
OK
```


7. The host application writes a new value to “Transfer URL request” resource to notify to the server that a new transfer URL is requested:

```
AT#LWM2MSET=0,26299,0,31,0,123
```

```
OK
```

the ACK is then received:

```
#LWM2MMON: UPD, "/26299/0/31/0"
```

Tip: The resources of the same instances can be written all together also through the AT#LWM2MOBJSET command:

```
AT#LWM2MOBJSET=0,26299,0
```

```
{"bn":"/26299/0/",
```

```
"e":[
```

```
{ "n": "0", "v": 1 },
```

```
{ "n": "1", "sv": "debug_2022-01-01.log" },
```

```
{ "n": "3", "sv": "log" },
```

```
{ "n": "31", "v": 123 } ]
```

```
}
```

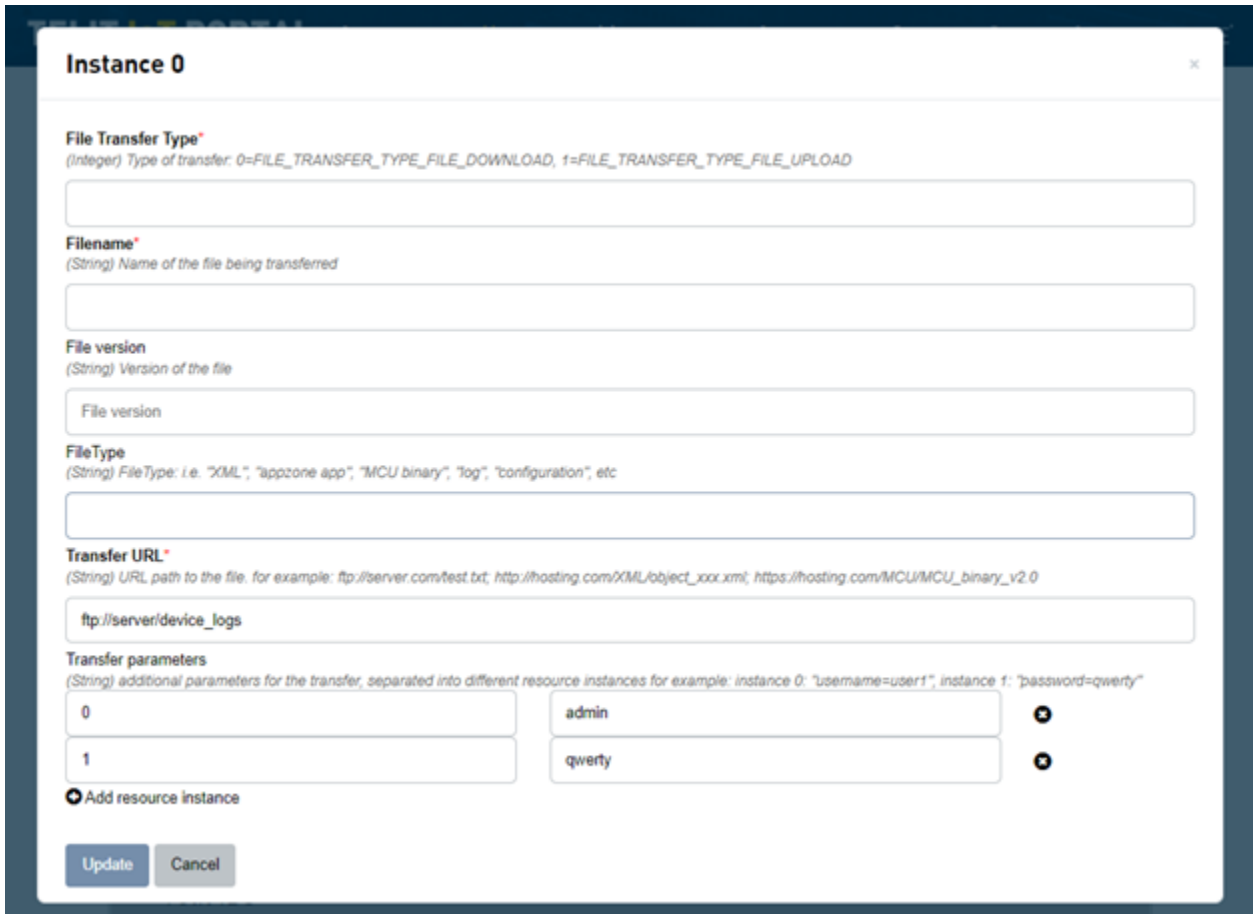
```
OK
```

8. On server side, the external application detects the new values and the “intelligent” agent retrieves the URL for the requested action.

The following steps are the same of the Use Case 1:

9. On server side, the external application writes the following values on the instance 0 of the “File Transfer Manager” object:

- Resource 11=“ftp://server/device_logs”
- Resource 12={0=“admin”, 1=“qwerty”}



Instance 0

File Transfer Type*
(Integer) Type of transfer: 0=FILE_TRANSFER_TYPE_FILE_DOWNLOAD, 1=FILE_TRANSFER_TYPE_FILE_UPLOAD

Filename*
(String) Name of the file being transferred

File version
(String) Version of the file

FileType
(String) FileType: i.e. "XML", "appzone app", "MCU binary", "log", "configuration", etc

Transfer URL*
(String) URL path to the file. for example: ftp://server.com/test.txt; http://hosting.com/XML/object_xxx.xml; https://hosting.com/MCU/MCU_binary_v2.0

Transfer parameters
(String) additional parameters for the transfer, separated into different resource instances for example: instance 0: "username=user1", instance 1: "password=qwerty"

0 admin

1 qwerty

+ Add resource instance

Update Cancel

Figure 14: writing multiple resources of instance 0 on OneEdge IoT portal

The same "write" operation can be done through the TR50 interface of OneEdge IoT portal:

```
{
  "data": {
    "command": "lwm2m.device.oir.write",
    "thingId": <thingId>,
    "objId": 26299,
    "instId": 0,
    "encoding": 11542,
    "params": {
      "data": {
        "11": "ftp:\\\\server\\device_logs",
        "12": ["admin","qwerty"]
      }
    }
  }
}
```

```

    }
  }
}
}

```

10. On modem side, the host application receives the following URC messages on the serial port:

```

#LWM2MMON: UPD, "/26299/0/11/0"
#LWM2MMON: UPD, "/26299/0/12/0"
#LWM2MMON: UPD, "/26299/0/12/1"

```

11. The host application reads the values of the resources:

```

AT#LWM2MR=0,26299,0,11,0
#LWM2MR: "ftp://server/device_logs"
OK
AT#LWM2MR=0,26299,0,12,0
#LWM2MR: "admin"
OK
AT#LWM2MR=0,26299,0,12,1
#LWM2MR: "qwerty"
OK

```

Tip: The resources of the same instances can be written all together also through the AT#LWM2MOBJGET command:

```

AT#LWM2MOBJGET=0,26299,0
{"bn":"/26299/0/",
 "e":[
  {"n":"0","v":1},
  {"n":"1","sv":"debug_2022-01-01.log"},
  {"n":"2","sv":""},
  {"n":"3","sv":"log"},
  {"n":"11","sv":"ftp:\\server\\device_logs"},
  {"n":"12/0","sv":"admin"},
  {"n":"12/1","sv":"qwerty"},
  ...
  {"n":"12/8","sv":""},
  {"n":"12/9","sv":""},
  {"n":"21","v":0},
  {"n":"22","sv":"loaded at factory"},
  {"n":"31","v":0}]
}
OK

```

12. The host application performs the file transfer according to the provided parameters: in this example, it's an **upload** (resource 0=1) to an **FTP** server with user "**admin**" and password "**qwerty**":

```
AT#SGACT=1,1
#SGACT: 10.4.21.156
OK
AT#FTPOPEN="server","admin","qwerty",1
OK
```

13. The host application could write the value "1" to "State" resource to notify to the server that the transfer is in progress:

```
AT#LWM2MSET=0,26299,0,21,0,1
OK
```

the ACK is then received:

```
#LWM2MMON: UPD, "/26299/0/21/0"
```

14. The host application performs the upload of the file:

```
AT#FTPCWD="device_logs"
OK
AT#FTPPUT="debug_2022_01_01.log",0
CONNECT
data here
+++
NO CARRIER
```

15. The host application writes the value "2" to "State" resource to notify to the server that the transfer is completed:

```
AT#LWM2MSET=0,26299,0,21,0,2
OK
```

the ACK is then received:

```
#LWM2MMON: UPD, "/26299/0/21/0"
```

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- it can interfere with other electronic devices, particularly in environments such as hospitals, airports, aircrafts, etc.
- there is a risk of explosion such as gasoline stations, oil refineries, etc. It is the responsibility of the user to enforce the country regulation and the specific environment regulation.

Do not disassemble the product; any mark of tampering will compromise the warranty validity. We recommend following the instructions of the hardware user guides for correct wiring of the product. The product has to be supplied with a stabilized voltage source and the wiring has to be conformed to the security and fire prevention regulations. The product has to be handled with care, avoiding any contact with the pins because electrostatic discharges may damage the product itself. Same cautions have to be taken for the SIM, checking carefully the instruction for its use. Do not insert or remove the SIM when the product is in power saving mode.

The system integrator is responsible for the functioning of the final product. Therefore, the external components of the module, as well as any project or installation issue, have to be handled with care. Any interference may cause the risk of disturbing the GSM network or external devices or having an impact on the security system. Should there be any doubt, please refer to the technical documentation and the regulations in force. Every module has to be equipped with a proper antenna with specific characteristics. The antenna has to be installed carefully in order to avoid any interference with other electronic devices and has to guarantee a minimum distance from the body (20 cm). In case this requirement cannot be satisfied, the system integrator has to assess the final product against the SAR regulation.

The equipment is intended to be installed in a restricted area location.

The equipment must be supplied by an external specific limited power source in compliance with the standard EN 62368-1:2014.

The European Community provides some Directives for the electronic equipment introduced on the market. All of the relevant information is available on the European Community website:



https://ec.europa.eu/growth/sectors/electrical-engineering_en

7. GLOSSARY

ADC	Analog – Digital Converter
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
GPIO	General Purpose Input Output
HS	High Speed
HSDPA	High Speed Downlink Packet Access
HSIC	High Speed Inter Chip
HSUPA	High Speed Uplink Packet Access
I/O	Input Output
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
PCB	Printed Circuit Board
RTC	Real Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
USB	Universal Serial Bus
VNA	Vector Network Analyzer
VSWR	Voltage Standing Wave Ratio
WCDMA	Wideband Code Division Multiple Access



8. DOCUMENT HISTORY

Revision	Date	Changes
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0	2022-03-24	First issue

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